МОСКОВСКИЙ АВИАЦИОННЫЙ ИНСТИТУТ
(НАЦИОНАЛЬНЫЙ ИССЛЕДОВАТЕЛЬСКИЙ УНИВЕРСИТЕТ)

Институт №8 «Компьютерные науки и прикладная математика»
Кафедра 806 «Вычислительная математика и программирование»

Лабораторная работа № 3 по курсу «Операционные системы и архитектура компьютера»

Выполнил: Новак Дмитрий Евгеньевич
Группа: М8О-201БВ-25
Преподаватель: Миронов Евгений Сергеевич
Дата: 15.10.2026
Оценка: _____

Москва, 2026

Условие

Цель работы: получить практические навыки создания потоков и синхронизации между ними, исследовать ускорение параллельного алгоритма.

Задание: написать на языке C многопоточную программу. Максимальное количество одновременно работающих потоков задаётся ключом запуска. Нужно показать число потоков средствами ОС и исследовать влияние размера входных данных и числа потоков на ускорение и эффективность.

Вариант 3: сортировка массива целых чисел параллельным алгоритмом слияния.

Метод решения

Программа читает целые числа из файла в массив. Сортировка выполняется итеративно: на проходе с шириной w сливаются соседние отсортированные отрезки длиной не более w, затем w удваивается. Каждый поток получает независимые пары отрезков по циклическому распределению. Потоки читают из одного массива и записывают в непересекающиеся участки второго. После завершения всех слияний прохода барьер позволяет одному потоку поменять указатели на массивы. Второй барьер гарантирует, что остальные увидят новые указатели до следующего прохода.

Главный поток участвует в сортировке, поэтому при ключе `--threads p` создаётся не более $p - 1$ дополнительных потоков. Если элементов меньше p, используются только необходимые потоки. Алгоритм требует $O(n \log n)$ сравнений и $O(n)$ дополнительной памяти. Слияние на одном проходе делится между потоками без блокировок на элементах массива. На последних проходах пар мало, поэтому число одновременно занятых вычислениями потоков может быть меньше заданного предела.

Описание программы

Структура файлов

Таблица 1 — Основные файлы лабораторной работы

Файл	Назначение
`src/merge_sort.c`	Чтение, параллельная сортировка и вывод результата.
`CMakeLists.txt`	Сборка программы и подключение POSIX Threads.
`data/numbers_*.txt`	Входные массивы из 100, 1 000, 10 000, 100 000 и 1 000 000 чисел.
`report/measurements.csv`	Результаты измерений.
`report/assets/*.png`	Графики ускорения и эффективности.
`report/trace/*.txt`	Краткая и полная трассировки запуска в Ubuntu.

Основные типы данных и функции

`SortState` хранит исходный и вспомогательный массивы, длину, число потоков и барьер. `Worker` задаёт идентификатор потока и ссылку на общее состояние. `Barrier` построен на мьютексе и условной переменной; счётчик поколений защищает от ложных пробуждений.

`read_numbers()` читает и проверяет числа типа `int`.

`merge_block()` сливает одну пару отрезков. `sort_worker()` выполняет проходы сортировки. `sort_numbers()` создаёт потоки, дожидается их завершения и возвращает указатель на отсортированный массив.

Используемые системные вызовы и POSIX-интерфейсы

Таблица 2 — Интерфейсы ОС, используемые программой

Интерфейс	Назначение
`pthread_create()`	Создание дополнительных потоков.
`pthread_join()`	Ожидание завершения потоков.
`pthread_mutex_*()`	Взаимное исключение при изменении состояния барьера.
`pthread_cond_*()`	Ожидание старта и окончания прохода; пробуждение потоков.
`clock_gettime()`	Измерение времени монотонными часами.
`getpid()`	Вывод PID для проверки числа потоков средствами ОС.
`fopen()`, `getline()`, `fclose()`	Чтение входного файла через стандартную библиотеку C.

`pthread_*()` и функции стандартной библиотеки являются пользовательскими интерфейсами; их реализация может вызывать разные системные вызовы в Linux и macOS. Фактические вызовы Linux следует смотреть в трассировке `strace`.

Ключевые фрагменты кода

Мьютекс защищает счётчик пришедших потоков. Последний поток меняет поколение барьера и пробуждает остальные через условную переменную.

Листинг 1 — Барьер между проходами (merge_sort.c)

```
void wait_at_barrier(Barrier* barrier) {
    check_pthread(pthread_mutex_lock(&barrier->mutex),
     ↪  "pthread_mutex_lock");
    size_t generation = barrier->generation;
    barrier->arrived++;

    if (barrier->arrived == barrier->participants) {
        barrier->arrived = 0;
        barrier->generation++;
        check_pthread(pthread_cond_broadcast(&barrier->condition),
         ↪  "pthread_cond_broadcast");
    } else {
        while (generation == barrier->generation) {
            check_pthread(pthread_cond_wait(&barrier->condition,
             ↪  &barrier->mutex), "pthread_cond_wait");
        }
    }
```

```
    check_pthread(pthread_mutex_unlock(&barrier->mutex),
     ↪  "pthread_mutex_unlock");
}
```

Каждый поток получает номера пар с шагом, равным числу потоков. Одна пара обрабатывается только одним потоком.

Листинг 2 — Распределение блоков и синхронизация (merge_sort.c)

```
    for (size_t width = 1; width < state->length; width *= 2) {
        size_t span = width * 2;
        size_t blocks = state->length / span + (state->length %
         ↪  span != 0);

        for (size_t block = worker->id; block < blocks; block +=
         ↪  state->thread_count) {
            merge_block(state, block * span, width);
        }

        wait_at_barrier(&state->barrier);

        if (worker->id == 0) {
            int* previous = state->source;
            state->source = state->destination;
            state->destination = previous;
        }

        wait_at_barrier(&state->barrier);
```

При создании очередного потока проверяется код ошибки. Если создание не удалось, уже созданные потоки освобождаются из ожидания и завершаются.

Листинг 3 — Создание потоков (merge_sort.c)

```
    for (size_t id = 1; id < *used_threads; id++) {
        workers[id].state = &state;
        workers[id].id = id;

        int error = pthread_create(&threads[id - 1], NULL,
         ↪  sort_worker, &workers[id]);

        if (error != 0) {
            fprintf(stderr, "pthread_create: %s\n",
             ↪  strerror(error));

            check_pthread(pthread_mutex_lock(&state.barrier.mutex)⌋
             ↪  ,
             ↪  "pthread_mutex_lock");
            state.barrier.cancelled = 1;
            state.barrier.started = 1;
            check_pthread(pthread_cond_broadcast(&state.barrier.co⌋
             ↪  ndition),
             ↪  "pthread_cond_broadcast");
```

```
            check_pthread(pthread_mutex_unlock(&state.barrier.mute⌋
             ↪  x),
             ↪  "pthread_mutex_unlock");

            for (size_t index = 0; index < created; index++) {
                check_pthread(pthread_join(threads[index], NULL),
                 ↪  "pthread_join");
            }

            check_pthread(pthread_cond_destroy(&state.barrier.cond⌋
             ↪  ition),
             ↪  "pthread_cond_destroy");
            check_pthread(pthread_mutex_destroy(&state.barrier.mut⌋
             ↪  ex),
             ↪  "pthread_mutex_destroy");
            free(buffer);
            free(threads);
            free(workers);
            return -1;
        }

        created++;
    }
```

Результаты

Программа собрана с флагами `-Wall -Wextra -Wpedantic`. Пример запуска на файле `lab_03/data/sample.txt`:

```
$ ./build/lab_03/lab3_merge_sort --threads 4 --input
 ↪  lab_03/data/sample.txt
-3 -1 0 4 5 5 8 8
```

Для исследования использованы пять входных файлов из каталога `data/`: 100, 1 000, 10 000, 100 000 и 1 000 000 целых чисел. Для каждого файла заданы пределы 1, 2, 4 и 8 потоков (включая главный). Измерения выполнены на macOS 26.6.2, ARM64, по пять запусков на каждую пару параметров; в таблице приведена медиана. Время T_{p} измеряется внутри программы по `CLOCK_MONOTONIC`: учитывается сортировка вместе с созданием и ожиданием потоков, чтение входного файла и печать массива не учитываются. Режим `--observe` при измерениях не использовался. Для каждого файла ускорение вычислено как $S_p = T_1/T_p$, эффективность — как $E_p = S_p/p$. Трассировка `strace` получена отдельно в Ubuntu и не является источником этих времён.

Таблица 3 — Время сортировки, ускорение и эффективность для входных файлов

n	p	T_p, мс	S_p	E_p
100	1	0.015	1.000	1.000
100	2	0.091	0.165	0.082
100	4	0.151	0.099	0.025
100	8	0.310	0.048	0.006
1 000	1	0.055	1.000	1.000
1 000	2	0.099	0.556	0.278
1 000	4	0.237	0.232	0.058
1 000	8	0.449	0.122	0.015
10 000	1	0.569	1.000	1.000
10 000	2	0.444	1.282	0.641
10 000	4	0.396	1.437	0.359
10 000	8	0.757	0.752	0.094
100 000	1	6.824	1.000	1.000
100 000	2	4.170	1.636	0.818
100 000	4	2.637	2.588	0.647
100 000	8	2.846	2.398	0.300
1 000 000	1	82.537	1.000	1.000
1 000 000	2	45.829	1.801	0.900
1 000 000	4	27.440	3.008	0.752
1 000 000	8	23.370	3.532	0.441

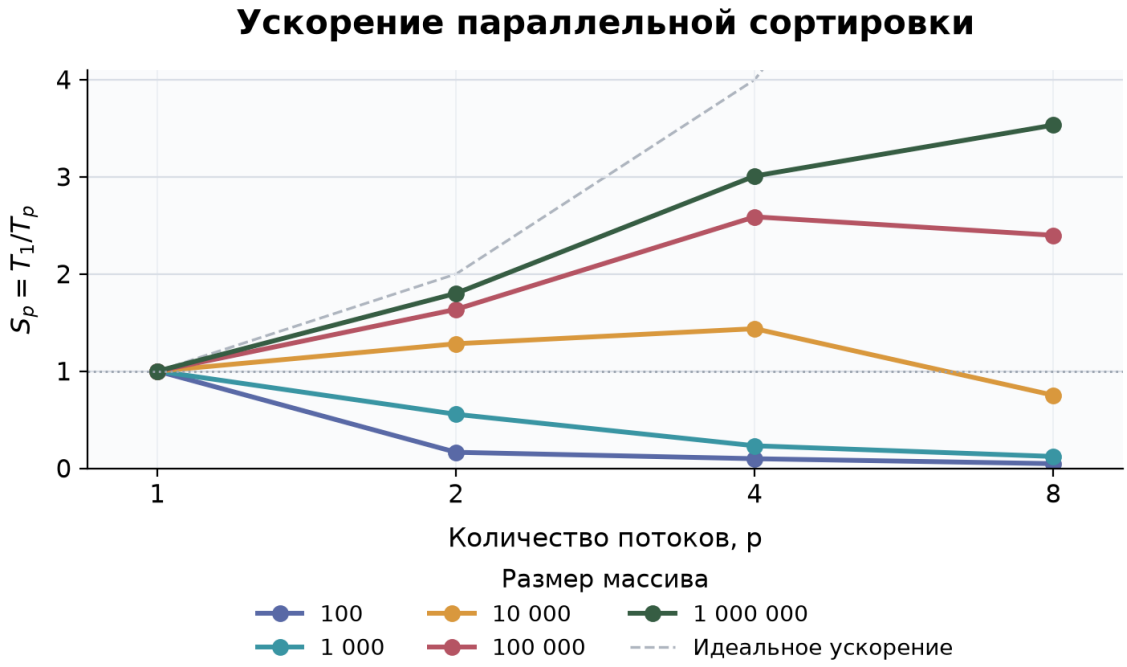


Рисунок 1 — Зависимость ускорения от числа потоков для пяти входных файлов

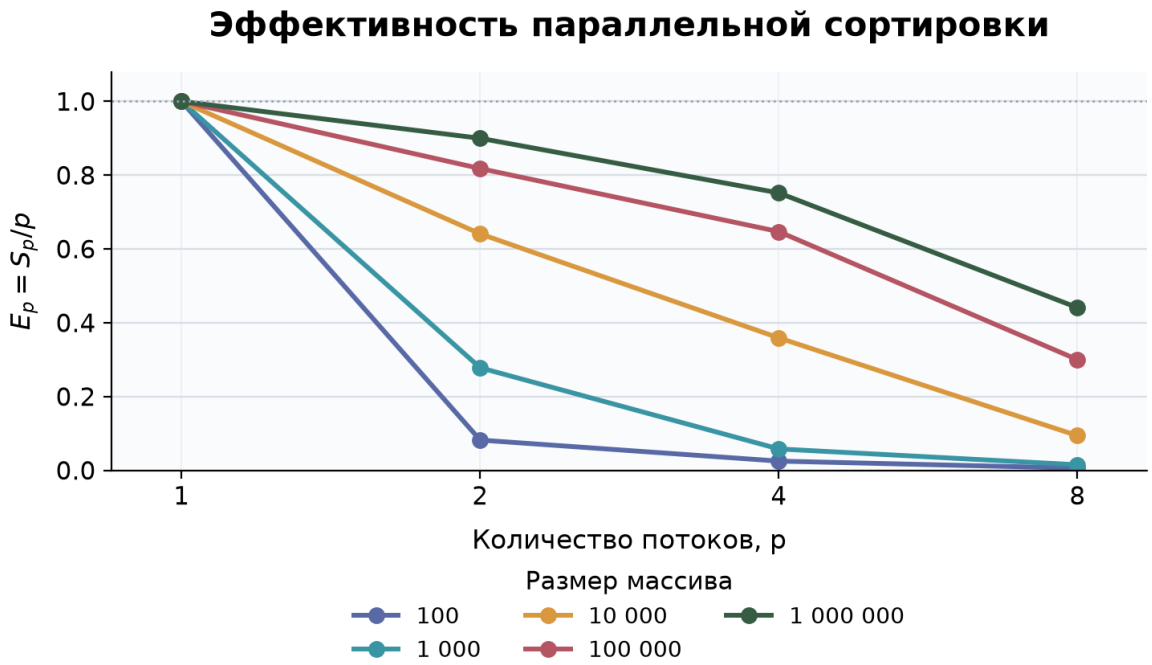


Рисунок 2 — Зависимость эффективности от числа потоков для пяти входных файлов

На массивах из 100 и 1 000 элементов один поток работает быстрее нескольких. Для 100 чисел время при одном потоке составило 0.015 мс, при восьми — 0.310 мс. Объём полезной работы мал, а создание потоков и два барьера на каждом проходе требуют времени. Для файла на 10 000 чисел лучший результат дали четыре потока; ускорение составило $S_4 = 1.437$. Для файла на 100 000 чисел тоже лучше оказались четыре потока: $S_4 = 2.588$. При восьми потоках на этих размерах ускорение уменьшается из-за накладных расходов.

На массиве из 1 000 000 элементов время сократилось с 82.537 мс при одном потоке до 23.370 мс при восьми: $S_8 = 3.532$, $E_8 = 0.441$. Эффективность снижается с ростом числа потоков, поскольку на последних проходах остаётся мало независимых слияний; потоки ожидают друг друга на барьерах. На результаты также влияют затраты планировщика и пропускная способность памяти. Полные численные данные сохранены в `report/measurements.csv` и приведены в приложении Б.

Демонстрация работы strace

Трассировка получена в Ubuntu ARM64 для файла `numbers_10000.txt` с ключом `--threads 4`. Команда запуска из корня репозитория:

```
$ strace -f -o lab_03/report/trace/strace.txt
↪  ./build/lab_03/lab3_merge_sort --threads 4 --input
↪  lab_03/data/numbers_10000.txt > /dev/null
```

Ниже приведены строки из сокращённой трассировки; полный вывод находится в `report/trace/strace.txt` и приложении В.

Листинг 4 — Ключевые системные вызовы из strace

```
821490 execve("./build/lab_03/lab3_merge_sort",
↪  ["./build/lab_03/lab3_merge_sort", "--threads", "4", "--input",
↪  "lab_03/data/numbers_10000.txt"], 0xfffff06d94d8 /* 65 vars */) = 0
821490 openat(AT_FDCWD, "lab_03/data/numbers_10000.txt", O_RDONLY) = 3
821490 read(3, "917515 -556988 -583845 -76703 86"..., 4096) = 4096
821490 clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_T⌋
↪  HREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLE⌋
↪  ARTID, child_tid=0xf7b137f1f588, parent_tid=0xf7b137f1f230,
↪  exit_signal=0, stack=0xf7b137710000, stack_size=0x80ea40,
↪  tls=0xf7b137f1f880} => {parent_tid=[821491]}, 88) = 821491
821490 futex(0xffffc605db28, FUTEX_WAKE_PRIVATE, 2147483647) = 1
821490 futex(0xffffc605db28, FUTEX_WAKE_PRIVATE, 2147483647) = 3
821490 futex(0xf7b137f1f588, FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 1,
↪  NULL, FUTEX_BITSET_MATCH_ANY <unfinished ...>
821490 <... futex resumed>)             = 0
821490 exit_group(0)                    = ?
```

Вызов `openat()` открывает входной файл, а `read()` считывает числа. В трассировке видны три создания потока через `clone3()`: главный поток 821490 и дополнительные потоки 821491–821493 вместе дают четыре потока. Флаги `CLONE_VM` и `CLONE_THREAD` показывают общее адресное пространство и принадлежность к одной группе потоков.

Ожидание и пробуждение при работе мьютекса и условной переменной наблюдаются как вызовы `futex()`. После завершения дополнительных потоков главный выводит время и отсортированный массив, затем вызывает `exit_group(0)`. На уровне исходного кода эти действия выполняются через интерфейсы `pthread_*()` и стандартной библиотеки C.

Выводы

Таким образом, данная лабораторная работа никак кардинально не изменила мою жизнь: я написал свою первую многопоточную программу и благодаря лабораторной понял, что можно параллелить разные задачи. Всегда было интересно, как параллелят разные задачи для алгоритмов машинного обучения (очень трудно следить, чтобы все потоки правильно отработали, а потом объединили свои усилия в бОльшую задачу), а тут у самого получилось написать работающую сортировку. Очень трудоемко.

Приложение А. Исходный код программы

Листинг А.1 — Файл merge_sort.c

```
#include <ctype.h>
#include <errno.h>
#include <limits.h>
#include <pthread.h>
#include <stdint.h>
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <time.h>
#include <unistd.h>

#define GENERAL_ERROR_EXIT_CODE 1
#define INPUT_ERROR_EXIT_CODE 2

typedef struct {
    pthread_mutex_t mutex;
    pthread_cond_t condition;
    size_t arrived;
    size_t generation;
    size_t participants;
    int started;
    int cancelled;
} Barrier;

typedef struct {
    int* source;
    int* destination;
    size_t length;
    size_t thread_count;
    Barrier barrier;
} SortState;

typedef struct {
    SortState* state;
    size_t id;
} Worker;

int read_numbers(const char* filename, int** numbers, size_t* length) {
    FILE* input = fopen(filename, "r");

    if (input == NULL) {
        perror("fopen");
        return GENERAL_ERROR_EXIT_CODE;
    }

    char* line = NULL;
    size_t line_capacity = 0;
    size_t capacity = 0;
    *numbers = NULL;
    *length = 0;

    while (getline(&line, &line_capacity, input) != -1) {
        char* current = line;

        while (*current != '\0') {
            while (isspace((unsigned char)*current)) {
                current++;
            }

            if (*current == '\0') {
```

```
            break;
        }

        char* end = NULL;
        errno = 0;
        long value = strtol(current, &end, 10);

        if (current == end || errno == ERANGE || value < INT_MIN ||
        ↪  value > INT_MAX) {
            fprintf(stderr, "Ошибка: входной файл содержит
            ↪  некорректное целое число.\n");
            free(line);
            fclose(input);
            free(*numbers);
            *numbers = NULL;
            return INPUT_ERROR_EXIT_CODE;
        }

        if (*length == capacity) {
            size_t next_capacity = capacity == 0 ? 1024 : capacity *
            ↪  2;

            if (next_capacity < capacity || next_capacity > SIZE_MAX
            ↪  / sizeof(int)) {
                fprintf(stderr, "Ошибка: входной файл слишком
                ↪  велик.\n");
                free(line);
                fclose(input);
                free(*numbers);
                *numbers = NULL;
                return GENERAL_ERROR_EXIT_CODE;
            }

            int* resized = realloc(*numbers, next_capacity *
            ↪  sizeof(int));

            if (resized == NULL) {
                perror("realloc");
                free(line);
                fclose(input);
                free(*numbers);
                *numbers = NULL;
                return GENERAL_ERROR_EXIT_CODE;
            }

            *numbers = resized;
            capacity = next_capacity;
        }

        (*numbers)[(*length)++] = (int)value;
        current = end;
    }
}

int read_error = ferror(input);
free(line);

int close_error = fclose(input);

if (read_error || close_error == EOF) {
    perror(read_error ? "getline" : "fclose");
    free(*numbers);
    *numbers = NULL;
    return GENERAL_ERROR_EXIT_CODE;
```

```
    }

    return 0;
}

void check_pthread(int error, const char* function) {
    if (error != 0) {
        fprintf(stderr, "%s: %s\n", function, strerror(error));
        exit(GENERAL_ERROR_EXIT_CODE);
    }
}

int init_barrier(Barrier* barrier, size_t participants) {
    int error = pthread_mutex_init(&barrier->mutex, NULL);

    if (error != 0) {
        fprintf(stderr, "pthread_mutex_init: %s\n", strerror(error));
        return -1;
    }

    error = pthread_cond_init(&barrier->condition, NULL);

    if (error != 0) {
        fprintf(stderr, "pthread_cond_init: %s\n", strerror(error));
        check_pthread(pthread_mutex_destroy(&barrier->mutex),
        ↪  "pthread_mutex_destroy");
        return -1;
    }

    barrier->arrived = 0;
    barrier->generation = 0;
    barrier->participants = participants;
    barrier->started = 0;
    barrier->cancelled = 0;
    return 0;
}

void wait_for_start(Barrier* barrier) {
    check_pthread(pthread_mutex_lock(&barrier->mutex),
    ↪  "pthread_mutex_lock");

    while (!barrier->started) {
        check_pthread(pthread_cond_wait(&barrier->condition,
        ↪  &barrier->mutex), "pthread_cond_wait");
    }

    check_pthread(pthread_mutex_unlock(&barrier->mutex),
    ↪  "pthread_mutex_unlock");
}

void wait_at_barrier(Barrier* barrier) {
    check_pthread(pthread_mutex_lock(&barrier->mutex),
    ↪  "pthread_mutex_lock");
    size_t generation = barrier->generation;
    barrier->arrived++;

    if (barrier->arrived == barrier->participants) {
        barrier->arrived = 0;
        barrier->generation++;
        check_pthread(pthread_cond_broadcast(&barrier->condition),
        ↪  "pthread_cond_broadcast");
    } else {
        while (generation == barrier->generation) {
            check_pthread(pthread_cond_wait(&barrier->condition,
            ↪  &barrier->mutex), "pthread_cond_wait");
```

```
        }
    }

    check_pthread(pthread_mutex_unlock(&barrier->mutex),
    ↪  "pthread_mutex_unlock");
}

void merge_block(SortState* state, size_t left, size_t width) {
    size_t span = width * 2;
    size_t middle = left + width < state->length ? left + width :
    ↪  state->length;
    size_t right = left + span < state->length ? left + span :
    ↪  state->length;
    size_t first = left;
    size_t second = middle;

    for (size_t output = left; output < right; output++) {
        if (second == right || (first < middle && state->source[first]
        ↪  <= state->source[second])) {
            state->destination[output] = state->source[first++];
        } else {
            state->destination[output] = state->source[second++];
        }
    }
}

void* sort_worker(void* argument) {
    Worker* worker = argument;
    SortState* state = worker->state;
    wait_for_start(&state->barrier);

    if (state->barrier.cancelled) {
        return NULL;
    }

    for (size_t width = 1; width < state->length; width *= 2) {
        size_t span = width * 2;
        size_t blocks = state->length / span + (state->length % span !=
        ↪  0);

        for (size_t block = worker->id; block < blocks; block +=
        ↪  state->thread_count) {
            merge_block(state, block * span, width);
        }

        wait_at_barrier(&state->barrier);

        if (worker->id == 0) {
            int* previous = state->source;
            state->source = state->destination;
            state->destination = previous;
        }

        wait_at_barrier(&state->barrier);
    }

    return NULL;
}

int sort_numbers(int* numbers, size_t length, size_t limit, int observe,
↪  int** result, size_t* used_threads) {
    if (length == 0) {
        *result = numbers;
        *used_threads = 1;
```

```
        return 0;
    }

    *used_threads = length < limit ? length : limit;

    int* buffer = malloc(length * sizeof(int));
    pthread_t* threads = NULL;

    if (*used_threads > 1) {
        threads = calloc(*used_threads - 1, sizeof(pthread_t));
    }

    Worker* workers = calloc(*used_threads, sizeof(Worker));

    if (buffer == NULL || workers == NULL || (*used_threads > 1 &&
    ↪  threads == NULL)) {
        perror("malloc/calloc");
        free(buffer);
        free(threads);
        free(workers);
        return -1;
    }

    SortState state;
    state.source = numbers;
    state.destination = buffer;
    state.length = length;
    state.thread_count = *used_threads;

    if (init_barrier(&state.barrier, *used_threads) != 0) {
        free(buffer);
        free(threads);
        free(workers);
        return -1;
    }

    workers[0].state = &state;
    workers[0].id = 0;
    size_t created = 0;

    for (size_t id = 1; id < *used_threads; id++) {
        workers[id].state = &state;
        workers[id].id = id;

        int error = pthread_create(&threads[id - 1], NULL, sort_worker,
        ↪  &workers[id]);

        if (error != 0) {
            fprintf(stderr, "pthread_create: %s\n", strerror(error));

            check_pthread(pthread_mutex_lock(&state.barrier.mutex),
            ↪  "pthread_mutex_lock");
            state.barrier.cancelled = 1;
            state.barrier.started = 1;
            check_pthread(pthread_cond_broadcast(&state.barrier.conditio⌋
            ↪  n),
            ↪  "pthread_cond_broadcast");
            check_pthread(pthread_mutex_unlock(&state.barrier.mutex),
            ↪  "pthread_mutex_unlock");

            for (size_t index = 0; index < created; index++) {
                check_pthread(pthread_join(threads[index], NULL),
                ↪  "pthread_join");
            }
```

```
            check_pthread(pthread_cond_destroy(&state.barrier.condition)
            ↪  ,
            ↪  "pthread_cond_destroy");
            check_pthread(pthread_mutex_destroy(&state.barrier.mutex),
            ↪  "pthread_mutex_destroy");
            free(buffer);
            free(threads);
            free(workers);
            return -1;
        }

        created++;
    }

    if (observe) {
        fprintf(stderr, "PID %ld: всего %zu потоков (включая главный).
        ↪  Нажмите Enter для сортировки...\n", (long)getpid(),
        ↪  *used_threads);
        getchar();
    }

    check_pthread(pthread_mutex_lock(&state.barrier.mutex),
    ↪  "pthread_mutex_lock");
    state.barrier.started = 1;
    check_pthread(pthread_cond_broadcast(&state.barrier.condition),
    ↪  "pthread_cond_broadcast");
    check_pthread(pthread_mutex_unlock(&state.barrier.mutex),
    ↪  "pthread_mutex_unlock");

    sort_worker(&workers[0]);

    for (size_t index = 0; index < created; index++) {
        check_pthread(pthread_join(threads[index], NULL),
        ↪  "pthread_join");
    }

    check_pthread(pthread_cond_destroy(&state.barrier.condition),
    ↪  "pthread_cond_destroy");
    check_pthread(pthread_mutex_destroy(&state.barrier.mutex),
    ↪  "pthread_mutex_destroy");
    free(threads);
    free(workers);

    if (state.source == numbers) {
        free(buffer);
    } else {
        free(numbers);
    }

    *result = state.source;
    return 0;
}

int main(int argc, char** argv) {
    const char* filename = NULL;
    size_t thread_limit = 0;
    int observe = 0;

    for (int index = 1; index < argc; index++) {
        if (strcmp(argv[index], "--threads") == 0 && index + 1 < argc &&
        ↪  thread_limit == 0) {
            char* end = NULL;
            errno = 0;
```

```
            unsigned long value = strtoul(argv[++index], &end, 10);

            if (errno != 0 || end == argv[index] || *end != '\0' ||
            ↪  value == 0 || value > SIZE_MAX || argv[index][0] == '-')
            ↪  {
                fprintf(stderr, "Ошибка: количество потоков должно быть
                ↪  положительным целым числом.\n");
                return INPUT_ERROR_EXIT_CODE;
            }

            thread_limit = (size_t)value;
        } else if (strcmp(argv[index], "--input") == 0 && index + 1 <
        ↪  argc && filename == NULL) {
            filename = argv[++index];
        } else if (strcmp(argv[index], "--observe") == 0 && !observe) {
            observe = 1;
        } else {
            fprintf(stderr, "Ошибка: неизвестный или повторный
            ↪  аргумент.\n");
            return INPUT_ERROR_EXIT_CODE;
        }
    }

    if (thread_limit == 0 || filename == NULL) {
        fprintf(stderr, "Использование: %s --threads N --input FILE
        ↪  [--observe]\n", argv[0]);
        return INPUT_ERROR_EXIT_CODE;
    }

    int* numbers = NULL;
    size_t length = 0;
    int read_result = read_numbers(filename, &numbers, &length);

    if (read_result != 0) {
        return read_result;
    }

    if (length > SIZE_MAX / 2 / sizeof(int)) {
        fprintf(stderr, "Ошибка: массив слишком велик.\n");
        free(numbers);
        return GENERAL_ERROR_EXIT_CODE;
    }

    int* sorted = NULL;
    size_t used_threads = 0;

    struct timespec start;
    struct timespec finish;

    if (clock_gettime(CLOCK_MONOTONIC, &start) == -1) {
        perror("clock_gettime");
        free(numbers);
        return GENERAL_ERROR_EXIT_CODE;
    }

    if (sort_numbers(numbers, length, thread_limit, observe, &sorted,
    ↪  &used_threads) != 0) {
        free(numbers);
        return GENERAL_ERROR_EXIT_CODE;
    }

    if (clock_gettime(CLOCK_MONOTONIC, &finish) == -1) {
        perror("clock_gettime");
        free(sorted);
```

```
        return GENERAL_ERROR_EXIT_CODE;
    }

    double milliseconds = (double)(finish.tv_sec - start.tv_sec) *
    ↪  1000.0 +
                          (double)(finish.tv_nsec - start.tv_nsec) /
                          ↪  1000000.0;
    fprintf(stderr, "Время сортировки: %.3f мс; потоков: %zu\n",
    ↪  milliseconds, used_threads);

    for (size_t index = 0; index < length; index++) {
        printf("%d ", sorted[index]);
    }
    printf("\n");

    free(sorted);
    return 0;
}
```

Листинг А.2 — Файл CMakeLists.txt

```
cmake_minimum_required(VERSION 3.20)

project(lab_03_variant_3 C)

set(CMAKE_C_STANDARD 11)
set(CMAKE_C_STANDARD_REQUIRED ON)
set(CMAKE_C_EXTENSIONS ON)

find_package(Threads REQUIRED)
add_compile_options(-Wall -Wextra -Wpedantic)

add_executable(lab3_merge_sort src/merge_sort.c)
target_link_libraries(lab3_merge_sort PRIVATE Threads::Threads)
```

Приложение Б. Полные результаты измерений

Листинг Б.1 — Файл measurements.csv

```
input_file,elements,threads,median_ms,speedup,efficiency
numbers_100.txt,100,1,0.015,1.000,1.000
numbers_100.txt,100,2,0.091,0.165,0.082
numbers_100.txt,100,4,0.151,0.099,0.025
numbers_100.txt,100,8,0.310,0.048,0.006
numbers_1000.txt,1000,1,0.055,1.000,1.000
numbers_1000.txt,1000,2,0.099,0.556,0.278
numbers_1000.txt,1000,4,0.237,0.232,0.058
numbers_1000.txt,1000,8,0.449,0.122,0.015
numbers_10000.txt,10000,1,0.569,1.000,1.000
numbers_10000.txt,10000,2,0.444,1.282,0.641
numbers_10000.txt,10000,4,0.396,1.437,0.359
numbers_10000.txt,10000,8,0.757,0.752,0.094
numbers_100000.txt,100000,1,6.824,1.000,1.000
numbers_100000.txt,100000,2,4.170,1.636,0.818
numbers_100000.txt,100000,4,2.637,2.588,0.647
numbers_100000.txt,100000,8,2.846,2.398,0.300
numbers_1000000.txt,1000000,1,82.537,1.000,1.000
numbers_1000000.txt,1000000,2,45.829,1.801,0.900
numbers_1000000.txt,1000000,4,27.440,3.008,0.752
numbers_1000000.txt,1000000,8,23.370,3.532,0.441
```

Приложение B. Полный вывод strace

Листинг B.1 — Вывод strace -f

```
821490 execve("./build/lab_03/lab3_merge_sort",
 ↪  ["./build/lab_03/lab3_merge_sort", "--threads", "4", "--input",
 ↪  "lab_03/data/numbers_10000.txt"], 0xfffff06d94d8 /* 65 vars */) = 0
821490 brk(NULL)                        = 0xac1bb3a13000
821490 mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS,
 ↪  -1, 0) = 0xf7b138139000
821490 faccessat(AT_FDCWD, "/etc/ld.so.preload", R_OK) = -1 ENOENT (No
 ↪  such file or directory)
821490 openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
821490 fstat(3, {st_mode=S_IFREG|0644, st_size=63707, ...}) = 0
821490 mmap(NULL, 63707, PROT_READ, MAP_PRIVATE, 3, 0) = 0xf7b1380f3000
821490 close(3)                         = 0
821490 openat(AT_FDCWD, "/usr/lib/aarch64-linux-gnu/libc.so.6",
 ↪  O_RDONLY|O_CLOEXEC) = 3
821490 read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0\267\0\1\0\0\0\3101\2⌋
 ↪  \0\0\0\0\0"..., 832) =
 ↪  832
821490 fstat(3, {st_mode=S_IFREG|0755, st_size=1788240, ...}) = 0
821490 mmap(NULL, 1957440, PROT_NONE,
 ↪  MAP_PRIVATE|MAP_ANONYMOUS|MAP_DENYWRITE, -1, 0) = 0xf7b137f15000
821490 mmap(0xf7b137f20000, 1891904, PROT_READ|PROT_EXEC,
 ↪  MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0) = 0xf7b137f20000
821490 munmap(0xf7b137f15000, 45056)    = 0
821490 munmap(0xf7b1380ee000, 20032)    = 0
821490 mprotect(0xf7b1380c9000, 81920, PROT_NONE) = 0
821490 mmap(0xf7b1380dd000, 20480, PROT_READ|PROT_WRITE,
 ↪  MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1ad000) = 0xf7b1380dd000
821490 mmap(0xf7b1380e2000, 48704, PROT_READ|PROT_WRITE,
 ↪  MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0xf7b1380e2000
821490 mmap(0xf7b137f20000, 1740800, PROT_READ|PROT_EXEC|0x10,
 ↪  MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0) = 0xf7b137f20000
821490 close(3)                         = 0
821490 set_tid_address(0xf7b13813a388)  = 821490
821490 set_robust_list(0xf7b13813a040, 24) = 0
821490 rseq(0xf7b13813a740, 0x21, 0, 0xd428bc00) = 0
821490 mprotect(0xf7b1380dd000, 12288, PROT_READ) = 0
821490 mprotect(0xac1ba2c3f000, 4096, PROT_READ) = 0
821490 mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS,
 ↪  -1, 0) = 0xf7b138137000
821490 mprotect(0xf7b138141000, 8192, PROT_READ) = 0
821490 prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024,
 ↪  rlim_max=RLIM64_INFINITY}) = 0
821490 getrandom("\x00\x14\xa7\xb5\x50\x90\xda\xbf", 8, GRND_NONBLOCK) =
 ↪  8
821490 munmap(0xf7b1380f3000, 63707)    = 0
821490 brk(NULL)                        = 0xac1bb3a13000
821490 brk(0xac1bb3c00000)              = 0xac1bb3c00000
821490 openat(AT_FDCWD, "lab_03/data/numbers_10000.txt", O_RDONLY) = 3
821490 fstat(3, {st_mode=S_IFREG|0664, st_size=73897, ...}) = 0
821490 read(3, "917515 -556988 -583845 -76703 86"..., 4096) = 4096
821490 read(3, "599981 -973607 -31067 -678001 15"..., 4096) = 4096
821490 read(3, "1375 106861 602663 553477 -21513"..., 4096) = 4096
821490 read(3, "13439 -671615 113915 -703041 -18"..., 4096) = 4096
821490 read(3, "71045 113416 473928 756303 89554"..., 4096) = 4096
821490 read(3, "77 -833436 14898 -264477 36934 -"..., 4096) = 4096
```

```
821490 read(3, "7085 247737 472006 350946 -69389"..., 4096) = 4096
821490 read(3, "8 985097 -784577 -807372\n577060 "..., 4096) = 4096
821490 read(3, "734 -806398 -826714 722377 56378"..., 4096) = 4096
821490 read(3, "78732 388250 579570 -343622 -563"..., 4096) = 4096
821490 read(3, "594 832422 696725 -470075 -12793"..., 4096) = 4096
821490 read(3, "90 851093 301195 948110 -440917\n"..., 4096) = 4096
821490 read(3, "4451 -77667 962392 341788 668397"..., 4096) = 4096
821490 read(3, "38958 670021 -940413 893018 -892"..., 4096) = 4096
821490 read(3, "458 -146143 802998 333174 -22559"..., 4096) = 4096
821490 read(3, "3 565939 -23104 853897 -251011 -"..., 4096) = 4096
821490 read(3, " 827220 -616725 181430 771692 -7"..., 4096) = 4096
821490 read(3, "5 -198430 43389 475568 613517 56"..., 4096) = 4096
821490 read(3, " 668368 291123 -724895\n312195 36"..., 4096) = 169
821490 read(3, "", 4096)                = 0
821490 close(3)                         = 0
821490 rt_sigaction(SIGRT_1, {sa_handler=0xf7b137fa93c0, sa_mask=[],
↪  sa_flags=SA_ONSTACK|SA_RESTART|SA_SIGINFO}, NULL, 8) = 0
821490 rt_sigprocmask(SIG_UNBLOCK, [RTMIN RT_1], NULL, 8) = 0
821490 mmap(NULL, 8454144, PROT_READ|PROT_WRITE,
↪  MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0xf7b137710000
821490 madvise(0xf7b137710000, 65536, MADV_GUARD_INSTALL) = 0
821490 rt_sigprocmask(SIG_BLOCK, ~[], [], 8) = 0
821490 clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_T⌋
↪  HREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLE⌋
↪  ARTID, child_tid=0xf7b137f1f588, parent_tid=0xf7b137f1f230,
↪  exit_signal=0, stack=0xf7b137710000, stack_size=0x80ea40,
↪  tls=0xf7b137f1f880} => {parent_tid=[821491]}, 88) = 821491
821490 rt_sigprocmask(SIG_SETMASK, [], NULL, 8) = 0
821490 mmap(NULL, 8454144, PROT_READ|PROT_WRITE,
↪  MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0xf7b136f00000
821490 madvise(0xf7b136f00000, 65536, MADV_GUARD_INSTALL) = 0
821491 rseq(0xf7b137f1f940, 0x21, 0, 0xd428bc00 <unfinished ...>
821490 rt_sigprocmask(SIG_BLOCK, ~[] <unfinished ...>
821491 <... rseq resumed>)              = 0
821490 <... rt_sigprocmask resumed>, [], 8) = 0
821490 clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_T⌋
↪  HREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLE⌋
↪  ARTID, child_tid=0xf7b13770f588, parent_tid=0xf7b13770f230,
↪  exit_signal=0, stack=0xf7b136f00000, stack_size=0x80ea40,
↪  tls=0xf7b13770f880} <unfinished ...>
821491 set_robust_list(0xf7b137f1f240, 24 <unfinished ...>
821490 <... clone3 resumed> => {parent_tid=[821492]}, 88) = 821492
821491 <... set_robust_list resumed>)   = 0
821490 rt_sigprocmask(SIG_SETMASK, [], NULL, 8) = 0
821490 mmap(NULL, 8454144, PROT_READ|PROT_WRITE,
↪  MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0 <unfinished ...>
821491 rt_sigprocmask(SIG_SETMASK, [] <unfinished ...>
821490 <... mmap resumed>)              = 0xf7b1366f0000
821490 madvise(0xf7b1366f0000, 65536, MADV_GUARD_INSTALL) = 0
821490 rt_sigprocmask(SIG_BLOCK, ~[], [], 8) = 0
821490 clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_T⌋
↪  HREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLE⌋
↪  ARTID, child_tid=0xf7b136eff588, parent_tid=0xf7b136eff230,
↪  exit_signal=0, stack=0xf7b1366f0000, stack_size=0x80ea40,
↪  tls=0xf7b136eff880} <unfinished ...>
821491 <... rt_sigprocmask resumed>, NULL, 8) = 0
821490 <... clone3 resumed> => {parent_tid=[821493]}, 88) = 821493
```

```
821491 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 0, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821490 rt_sigprocmask(SIG_SETMASK, [] <unfinished ...>
821493 rseq(0xf7b136eff940, 0x21, 0, 0xd428bc00 <unfinished ...>
821490 <... rt_sigprocmask resumed>, NULL, 8) = 0
821490 futex(0xffffc605db28, FUTEX_WAKE_PRIVATE, 2147483647) = 1
821491 <... futex resumed>)             = 0
821490 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 0, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821493 <... rseq resumed>)              = 0
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 set_robust_list(0xf7b136eff240, 24 <unfinished ...>
821492 rseq(0xf7b13770f940, 0x21, 0, 0xd428bc00 <unfinished ...>
821491 <... futex resumed>)             = 0
821491 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 0, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821493 <... set_robust_list resumed>)   = 0
821492 <... rseq resumed>)              = 0
821493 rt_sigprocmask(SIG_SETMASK, [] <unfinished ...>
821492 set_robust_list(0xf7b13770f240, 24 <unfinished ...>
821493 <... rt_sigprocmask resumed>, NULL, 8) = 0
821492 <... set_robust_list resumed>)   = 0
821492 rt_sigprocmask(SIG_SETMASK, [] <unfinished ...>
821493 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 0, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821492 <... rt_sigprocmask resumed>, NULL, 8) = 0
821492 futex(0xffffc605db2c, FUTEX_WAKE_PRIVATE, 2147483647 <unfinished
↪  ...>
821490 <... futex resumed>)             = 0
821493 <... futex resumed>)             = 0
821492 <... futex resumed>)             = 3
821491 <... futex resumed>)             = 0
821490 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821491 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821492 <... futex resumed>)             = 1
821491 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821490 <... futex resumed>)             = 0
821490 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1) = 0
821490 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 0, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821492 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 0, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 <... futex resumed>)             = 0
821491 <... futex resumed>)             = 0
```

```
821493 futex(0xffffc605db28, FUTEX_WAKE_PRIVATE, 2147483647 <unfinished
↪  ...>
821491 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 0, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821490 <... futex resumed>)             = 0
821493 <... futex resumed>)             = 2
821492 <... futex resumed>)             = 0
821491 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821490 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821491 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821490 <... futex resumed>)             = 0
821493 <... futex resumed>)             = 0
821492 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821491 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 1, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821490 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 1, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821492 <... futex resumed>)             = 0
821491 <... futex resumed>)             = 0
821492 futex(0xffffc605db2c, FUTEX_WAKE_PRIVATE, 2147483647 <unfinished
↪  ...>
821491 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 1, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821493 <... futex resumed>)             = 0
821492 <... futex resumed>)             = 1
821491 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821493 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821492 <... futex resumed>)             = 0
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821491 <... futex resumed>)             = 0
821493 <... futex resumed>)             = 0
821492 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 4, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821493 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 4, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821491 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 4, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
```

```
821490 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821490 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1) = 0
821490 futex(0xffffc605db28, FUTEX_WAKE_PRIVATE, 2147483647) = 3
821493 <... futex resumed>)             = 0
821492 <... futex resumed>)             = 0
821491 <... futex resumed>)             = 0
821490 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 7, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 <... futex resumed>)             = 0
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 7, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821492 <... futex resumed>)             = 0
821491 <... futex resumed>)             = 0
821492 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 7, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821491 futex(0xffffc605db2c, FUTEX_WAKE_PRIVATE, 2147483647 <unfinished
↪  ...>
821492 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821490 <... futex resumed>)             = 0
821493 <... futex resumed>)             = 0
821491 <... futex resumed>)             = 2
821490 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821490 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821493 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821490 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821492 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821490 <... futex resumed>)             = 0
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821491 <... futex resumed>)             = 0
821490 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 10, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 <... futex resumed>)             = 0
821491 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 10, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821493 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 10, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821492 <... futex resumed>)             = 0
821492 futex(0xffffc605db28, FUTEX_WAKE_PRIVATE, 2147483647 <unfinished
↪  ...>
821490 <... futex resumed>)             = 0
```

```
821493 <... futex resumed>)             = 0
821492 <... futex resumed>)             = 3
821491 <... futex resumed>)             = 0
821490 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821490 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821493 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821490 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821492 <... futex resumed>)             = 0
821490 <... futex resumed>)             = 0
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821492 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 13, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821491 <... futex resumed>)             = 0
821490 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 13, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821493 <... futex resumed>)             = 0
821491 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 13, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821493 futex(0xffffc605db2c, FUTEX_WAKE_PRIVATE, 2147483647 <unfinished
↪  ...>
821491 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821490 <... futex resumed>)             = 0
821493 <... futex resumed>)             = 2
821492 <... futex resumed>)             = 0
821491 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821490 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821490 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821491 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821490 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 <... futex resumed>)             = 0
821492 <... futex resumed>)             = 0
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821490 <... futex resumed>)             = 0
821493 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 16, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821490 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 16, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821492 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 16, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821491 <... futex resumed>)             = 0
821491 futex(0xffffc605db28, FUTEX_WAKE_PRIVATE, 2147483647 <unfinished
↪  ...>
```

```
821490 <... futex resumed>)             = 0
821493 <... futex resumed>)             = 0
821492 <... futex resumed>)             = 0
821491 <... futex resumed>)             = 3
821490 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821490 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821493 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821492 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821490 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821491 <... futex resumed>)             = 0
821490 <... futex resumed>)             = 0
821492 <... futex resumed>)             = 0
821493 <... futex resumed>)             = 0
821491 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 19, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821490 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 19, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821493 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 19, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821490 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821493 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821492 futex(0xffffc605db2c, FUTEX_WAKE_PRIVATE, 2147483647 <unfinished
↪  ...>
821491 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821490 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821492 <... futex resumed>)             = 0
821491 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821490 <... futex resumed>)             = 0
821492 <... futex resumed>)             = 1
821490 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821492 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 22, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821490 <... futex resumed>)             = 1
821493 <... futex resumed>)             = 0
821490 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 22, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1) = 1
821491 <... futex resumed>)             = 0
```

```
821493 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 22, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1) = 0
821491 futex(0xffffc605db28, FUTEX_WAKE_PRIVATE, 2147483647 <unfinished
↪  ...>
821490 <... futex resumed>)             = 0
821493 <... futex resumed>)             = 0
821492 <... futex resumed>)             = 0
821491 <... futex resumed>)             = 3
821490 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821490 <... futex resumed>)             = 0
821493 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821492 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821491 <... futex resumed>)             = 0
821490 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 25, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821491 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 25, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821492 <... futex resumed>)             = 0
821493 <... futex resumed>)             = 0
821492 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 25, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821493 futex(0xffffc605db2c, FUTEX_WAKE_PRIVATE, 2147483647 <unfinished
↪  ...>
821490 <... futex resumed>)             = 0
821493 <... futex resumed>)             = 3
821492 <... futex resumed>)             = 0
821491 <... futex resumed>)             = 0
821490 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821490 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821493 <... futex resumed>)             = 0
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821490 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 28, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821492 <... futex resumed>)             = 0
821490 <... futex resumed>)             = 0
821492 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 28, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821491 <... futex resumed>)             = 0
```

```
821490 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 28, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821491 futex(0xffffc605db28, FUTEX_WAKE_PRIVATE, 2147483647 <unfinished
↪  ...>
821490 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821493 <... futex resumed>)             = 0
821492 <... futex resumed>)             = 0
821491 <... futex resumed>)             = 2
821490 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821490 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821493 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821492 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821491 <... futex resumed>)             = 0
821490 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821490 <... futex resumed>)             = 0
821493 <... futex resumed>)             = 0
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821491 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 31, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821490 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 31, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821492 <... futex resumed>)             = 0
821493 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 31, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821490 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821493 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821492 futex(0xffffc605db2c, FUTEX_WAKE_PRIVATE, 2147483647 <unfinished
↪  ...>
821491 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821490 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821492 <... futex resumed>)             = 0
821491 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821490 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821490 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821492 <... futex resumed>)             = 1
821491 <... futex resumed>)             = 0
821490 <... futex resumed>)             = 1
821493 <... futex resumed>)             = 0
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
```

```
821492 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 34, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821490 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 34, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821491 <... futex resumed>)             = 0
821493 <... futex resumed>)             = 0
821491 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 34, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821493 futex(0xffffc605db28, FUTEX_WAKE_PRIVATE, 2147483647 <unfinished
↪  ...>
821491 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821490 <... futex resumed>)             = 0
821493 <... futex resumed>)             = 2
821492 <... futex resumed>)             = 0
821491 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821490 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821490 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821493 <... futex resumed>)             = 1
821492 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821491 <... futex resumed>)             = 0
821490 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821490 <... futex resumed>)             = 0
821491 <... futex resumed>)             = 0
821493 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 37, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821492 <... futex resumed>)             = 0
821491 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 37, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821492 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 37, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821490 futex(0xffffc605db2c, FUTEX_WAKE_PRIVATE, 2147483647 <unfinished
↪  ...>
821493 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821491 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821492 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821490 <... futex resumed>)             = 0
821493 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821491 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821490 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1) = 1
821491 <... futex resumed>)             = 0
```

```
821490 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 40, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821492 <... futex resumed>)             = 0
821491 <... futex resumed>)             = 1
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821491 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 40, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821493 <... futex resumed>)             = 0
821492 <... futex resumed>)             = 1
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821492 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 40, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821493 <... futex resumed>)             = 0
821493 futex(0xffffc605db28, FUTEX_WAKE_PRIVATE, 2147483647 <unfinished
↪  ...>
821490 <... futex resumed>)             = 0
821493 <... futex resumed>)             = 3
821492 <... futex resumed>)             = 0
821491 <... futex resumed>)             = 0
821490 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821491 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821490 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821493 <... futex resumed>)             = 0
821492 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821491 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821490 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 43, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821490 <... futex resumed>)             = 0
821492 <... futex resumed>)             = 0
821491 <... futex resumed>)             = 0
821490 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 43, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821491 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 43, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821492 futex(0xffffc605db2c, FUTEX_WAKE_PRIVATE, 2147483647 <unfinished
↪  ...>
821490 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821491 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821490 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821493 <... futex resumed>)             = 0
821492 <... futex resumed>)             = 1
```

```
821491 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821490 <... futex resumed>)             = 0
821493 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821492 <... futex resumed>)             = 1
821490 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821492 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 46, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821490 <... futex resumed>)             = 1
821493 <... futex resumed>)             = 0
821491 <... futex resumed>)             = 0
821490 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 46, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821493 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 46, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1) = 0
821491 futex(0xffffc605db28, FUTEX_WAKE_PRIVATE, 2147483647 <unfinished
↪  ...>
821490 <... futex resumed>)             = 0
821493 <... futex resumed>)             = 0
821492 <... futex resumed>)             = 0
821491 <... futex resumed>)             = 3
821490 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821490 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821490 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821492 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821491 <... futex resumed>)             = 0
821490 <... futex resumed>)             = 0
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821491 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 49, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 <... futex resumed>)             = 0
821490 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 49, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821492 <... futex resumed>)             = 0
821493 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 49, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821490 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821492 futex(0xffffc605db2c, FUTEX_WAKE_PRIVATE, 2147483647 <unfinished
↪  ...>
```

```
821491 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821493 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821490 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821492 <... futex resumed>)             = 0
821491 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821490 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821490 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 <... futex resumed>)             = 0
821492 <... futex resumed>)             = 1
821490 <... futex resumed>)             = 1
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821491 <... futex resumed>)             = 0
821492 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 52, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821490 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 52, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821493 <... futex resumed>)             = 0
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 52, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821491 <... futex resumed>)             = 0
821491 futex(0xffffc605db28, FUTEX_WAKE_PRIVATE, 2147483647 <unfinished
↪  ...>
821490 <... futex resumed>)             = 0
821493 <... futex resumed>)             = 0
821492 <... futex resumed>)             = 0
821491 <... futex resumed>)             = 3
821490 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821490 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821493 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821492 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821491 <... futex resumed>)             = 0
821490 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1) = 0
821490 <... futex resumed>)             = 0
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821491 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 55, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821492 <... futex resumed>)             = 0
821493 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 55, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
```

```
821490 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 55, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821493 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821492 futex(0xffffc605db2c, FUTEX_WAKE_PRIVATE, 2147483647 <unfinished
↪  ...>
821491 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821490 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821493 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821492 <... futex resumed>)             = 0
821490 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821491 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821490 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821490 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821491 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821490 <... futex resumed>)             = 0
821493 <... futex resumed>)             = 0
821492 <... futex resumed>)             = 1
821490 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 58, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 <... futex resumed>)             = 0
821492 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 58, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821493 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 58, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821491 <... futex resumed>)             = 0
821491 futex(0xffffc605db28, FUTEX_WAKE_PRIVATE, 2147483647 <unfinished
↪  ...>
821490 <... futex resumed>)             = 0
821493 <... futex resumed>)             = 0
821492 <... futex resumed>)             = 0
821491 <... futex resumed>)             = 3
821490 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821490 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821493 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821492 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821491 <... futex resumed>)             = 0
821490 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
```

```
821491 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 61, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821490 <... futex resumed>)             = 0
821493 <... futex resumed>)             = 0
821492 <... futex resumed>)             = 0
821493 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 61, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821492 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 61, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821490 futex(0xffffc605db2c, FUTEX_WAKE_PRIVATE, 2147483647) = 1
821493 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821492 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821491 <... futex resumed>)             = 0
821490 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 64, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 <... futex resumed>)             = 0
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 64, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821492 <... futex resumed>)             = 0
821491 <... futex resumed>)             = 0
821492 futex(0xffffc605db28, FUTEX_WAKE_PRIVATE, 2147483647 <unfinished
↪  ...>
821491 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 64, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821490 <... futex resumed>)             = 0
821493 <... futex resumed>)             = 0
821492 <... futex resumed>)             = 2
821491 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821490 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821491 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821490 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821490 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821492 <... futex resumed>)             = 0
821491 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821490 <... futex resumed>)             = 0
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821492 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 67, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
```

```
821493 <... futex resumed>)             = 0
821491 <... futex resumed>)             = 0
821490 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 67, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821493 futex(0xffffc605db2c, FUTEX_WAKE_PRIVATE, 2147483647 <unfinished
↪  ...>
821491 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 67, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821490 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821490 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821493 <... futex resumed>)             = 1
821492 <... futex resumed>)             = 0
821491 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821491 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821490 <... futex resumed>)             = 0
821493 <... futex resumed>)             = 1
821490 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 70, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821492 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821491 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821490 <... futex resumed>)             = 0
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821490 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 70, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821492 <... futex resumed>)             = 0
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821492 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 70, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821491 <... futex resumed>)             = 0
821491 futex(0xffffc605db28, FUTEX_WAKE_PRIVATE, 2147483647 <unfinished
↪  ...>
821490 <... futex resumed>)             = 0
821493 <... futex resumed>)             = 0
821492 <... futex resumed>)             = 0
821491 <... futex resumed>)             = 3
821490 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821490 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821493 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821492 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821491 <... futex resumed>)             = 0
```

```
821490 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821491 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 73, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821490 <... futex resumed>)             = 0
821493 <... futex resumed>)             = 0
821490 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 73, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821493 futex(0xffffc605db2c,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 73, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821492 <... futex resumed>)             = 0
821490 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821492 futex(0xffffc605db2c, FUTEX_WAKE_PRIVATE, 2147483647 <unfinished
↪  ...>
821490 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821493 <... futex resumed>)             = 0
821492 <... futex resumed>)             = 2
821491 <... futex resumed>)             = 0
821493 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821491 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821490 <... futex resumed>)             = 0
821493 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821492 <... futex resumed>)             = 1
821490 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821492 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 76, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821491 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821490 <... futex resumed>)             = 0
821493 <... futex resumed>)             = 0
821490 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 76, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 futex(0xffffc605db28,
↪  FUTEX_WAIT_BITSET_PRIVATE|FUTEX_CLOCK_REALTIME, 76, NULL,
↪  FUTEX_BITSET_MATCH_ANY <unfinished ...>
821491 <... futex resumed>)             = 0
821491 futex(0xffffc605db28, FUTEX_WAKE_PRIVATE, 2147483647 <unfinished
↪  ...>
821490 <... futex resumed>)             = 0
821493 <... futex resumed>)             = 0
821492 <... futex resumed>)             = 0
821491 <... futex resumed>)             = 3
821490 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAIT_PRIVATE, 2, NULL <unfinished ...>
821490 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
```

```
821493 <... futex resumed>)             = -1 EAGAIN (Resource
↪  temporarily unavailable)
821492 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821491 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821490 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821493 futex(0xffffc605dad8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
821492 <... futex resumed>)             = 0
821491 <... futex resumed>)             = 0
821492 rt_sigprocmask(SIG_BLOCK, ~[RT_1] <unfinished ...>
821491 rt_sigprocmask(SIG_BLOCK, ~[RT_1] <unfinished ...>
821490 <... futex resumed>)             = 0
821493 <... futex resumed>)             = 0
821492 <... rt_sigprocmask resumed>, NULL, 8) = 0
821490 futex(0xf7b137f1f588, FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 1,
↪  NULL, FUTEX_BITSET_MATCH_ANY <unfinished ...>
821493 rt_sigprocmask(SIG_BLOCK, ~[RT_1] <unfinished ...>
821491 <... rt_sigprocmask resumed>, NULL, 8) = 0
821493 <... rt_sigprocmask resumed>, NULL, 8) = 0
821492 madvise(0xf7b136f00000, 8314880, MADV_DONTNEED <unfinished ...>
821491 madvise(0xf7b137710000, 8314880, MADV_DONTNEED <unfinished ...>
821493 madvise(0xf7b1366f0000, 8314880, MADV_DONTNEED <unfinished ...>
821492 <... madvise resumed>)           = 0
821491 <... madvise resumed>)           = 0
821493 <... madvise resumed>)           = 0
821493 exit(0 <unfinished ...>
821492 exit(0 <unfinished ...>
821491 exit(0 <unfinished ...>
821493 <... exit resumed>)              = ?
821492 <... exit resumed>)              = ?
821491 <... exit resumed>)              = ?
821490 <... futex resumed>)             = 0
821493 +++ exited with 0 +++
821492 +++ exited with 0 +++
821491 +++ exited with 0 +++
821490 write(2, "\320\222\321\200\320\265\320\274\321\217
↪  \321\201\320\276\321\200\321\202\320\270\321\200\320\276\320\262\320⌋
↪  \272\320\270:"..., 64) =
↪  64
821490 fstat(1, {st_mode=S_IFCHR|0666, st_rdev=makedev(0x1, 0x3), ...})
↪  = 0
821490 ioctl(1, TCGETS2, 0xffffc605d738) = -1 ENOTTY (Inappropriate
↪  ioctl for device)
821490 write(1, "-999979 -999930 -999912 -999749 "..., 4096) = 4096
821490 write(1, "-907724 -907693 -907650 -907511 "..., 4096) = 4096
821490 write(1, "-811279 -811209 -811147 -811127 "..., 4096) = 4096
821490 write(1, "-709426 -709338 -709219 -709192 "..., 4096) = 4096
821490 write(1, "-607623 -607510 -607490 -607117 "..., 4096) = 4096
821490 write(1, "-501539 -501534 -501359 -501353 "..., 4096) = 4096
821490 write(1, "-398612 -398512 -398485 -398293 "..., 4096) = 4096
821490 write(1, "-290357 -290310 -290274 -290035 "..., 4096) = 4096
821490 write(1, "-185627 -185547 -185311 -185216 "..., 4096) = 4096
821490 write(1, "3 -78557 -78497 -78447 -78438 -7"..., 4096) = 4096
821490 write(1, "46036 46256 46362 46531 46796 46"..., 4096) = 4096
821490 write(1, " 165610 166079 166331 166494 166"..., 4096) = 4096
821490 write(1, "281605 281770 281803 281859 2819"..., 4096) = 4096
821490 write(1, "00715 400825 400922 401104 40199"..., 4096) = 4096
821490 write(1, "6321 526504 526677 527121 527138"..., 4096) = 4096
821490 write(1, "176 642887 642980 643066 643173 "..., 4096) = 4096
```

```
821490 write(1, "62 755410 755941 756025 756036 7"..., 4096) = 4096
821490 write(1, "4 870205 870621 870668 870799 87"..., 4096) = 4096
821490 write(1, " 995102 995247 995576 995714 995"..., 170) = 170
821490 exit_group(0)                    = ?
821490 +++ exited with 0 +++
```